TrackX AI - Software Requirements Specification (SRS)

AI-Based Smart Railway Track Defect Detection System | AI Synergy Hackathon 2026 | Team ORIONS

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1. Introduction

1.1 Purpose. This document specifies the requirements of TrackX AI, a prototype that uses ultrasonic testing, ESP32-based wireless nodes and machine learning to flag potential defects in steel rails and present results on a laptop dashboard.

1.2 Problem. Defects found late can grow into serious failures; manual inspection is slow and depends on human judgement; inspection is hard in rain, fog and at night; raw ultrasonic data takes expert time to analyse. 5,353 rail fractures were reported on Indian Railways in 2008-09 (Lok Sabha Unstarred Question No. 2, answered 19 Nov 2009 - an older official figure).

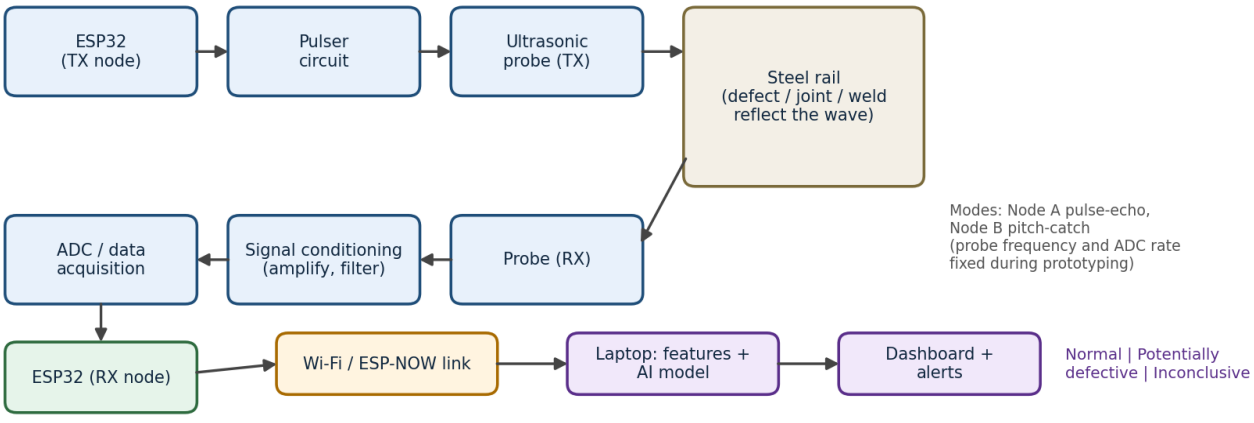
1.3 Scope and honesty statement. TrackX AI is a **prototype to assist inspection**. It is not a certified safety system. All data currently in this repository is **simulated**; real performance can only be shown with real labelled data and must be validated against established inspection methods before any railway safety use.

1.4 Definitions. TX/RX: transmit/receive node. ADC: analog-to-digital converter. SNR: signal-to-noise ratio. Inconclusive: output used when the signal is unreliable or the model is not confident.

2. System Overview

Two ESP32 nodes send and receive ultrasonic waves through the rail (Node A pulse-echo, Node B pitch-catch). A pulser-receiver, signal conditioning and ADC capture the waveform. The RX node sends it over Wi-Fi / ESP-NOW to a laptop, where features are extracted, an AI model classifies the waveform, and a dashboard shows results and alerts.

TrackX AI - System Architecture (block diagram)



3. Functional Requirements

ID	Requirement	Priority
FR-1	The TX node shall trigger the pulser circuit at a configurable interval.	Must

ID	Requirement	Priority
FR-2	The RX node shall digitise the received signal for a fixed capture window and transmit it to the laptop wirelessly.	Must
FR-3	The system shall extract features: echo amplitude, arrival time, energy, spectrum, SNR, attenuation. Raw waveforms shall also be stored.	Must
FR-4	The classifier shall output exactly one of: Normal, Potentially defective, Inconclusive.	Must
FR-5	A reading shall be Inconclusive when SNR is below a threshold, no echo is found, or class confidence is below a threshold.	Must
FR-6	The dashboard shall show the waveform, the class, confidence, SNR and a result log.	Must
FR-7	The dashboard shall raise an alert for Potentially defective and a re-measure prompt for Inconclusive.	Should
FR-8	The system shall store each waveform with sample ID, probe position, session and conditions.	Should
FR-9	The system should support a 1D-CNN model if enough labelled data exists.	Could
FR-10	Simulated data shall always be labelled as simulated (file names and metadata).	Must

4. Non-Functional Requirements

ID	Requirement
NFR-1	Classification of one waveform on a laptop shall complete in under 1 second (target).
NFR-2	Hardware shall be low-cost (planning range Rs 46,000 - 1.5 lakh; mid-point about Rs 98,000; rough ranges, not quotes).
NFR-3	Software shall use open-source tools (Python, scikit-learn, Streamlit).
NFR-4	The system shall not hide uncertain readings; they shall be flagged.
NFR-5	Reported metrics shall come from unseen samples or sessions only.

5. AI Requirements and Evaluation

Pipeline: data collection, dataset preparation, feature extraction, model training (Random Forest / SVM first; 1D-CNN if data allows), testing on unseen samples or sessions, deployment with the dashboard. Metrics: recall, precision, false-alarm rate, share of Inconclusive readings, and comparison with established inspection results. Challenges handled: joints/welds, temperature, vibration, noise and probe-coupling variation (location-wise data, temperature compensation, filtering).

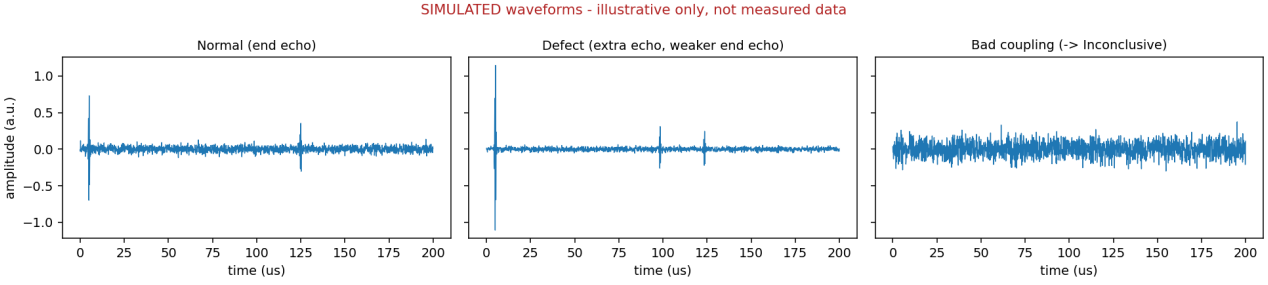


Figure: simulated waveforms, illustrative only, not measured data.

6. Constraints, Assumptions and Risks

Risk / challenge	Mitigation
Labelled defect data is hard to get	NDT lab and public datasets; simulated data only for early software work

Risk / challenge	Mitigation
Gaps, welds, temperature, vibration change the signal	Location-wise data, temperature compensation, filtering
Noise and probe-coupling variation	Signal conditioning and representative training examples
Weak, missing or unreliable signals	Detect them and request another reading (Inconclusive)
Wrong sensor choice (HC-SR04)	Proper ultrasonic probe with pulser-receiver electronics
ESP32 internal ADC too slow for MHz ultrasound	External high-speed ADC or analog front-end (to be fixed in prototyping)
Real-world safety validation	Validate against established inspection methods first

7. Roadmap (indicative)

Phase 1 (0-2 months) research and design; Phase 2 (2-5) prototype and data; Phase 3 (5-9) model and testing; Phase 4 (9-12+) dashboard and pilot. Milestones: design ready, data collected, model validated, pilot-ready.

8. References

[1] J. L. Rose, *Ultrasonic Guided Waves in Solid Media*, Cambridge University Press, 2014.

[2] J. L. Rose, M. J. Avioli and W. J. Song, "Application and potential of guided wave rail inspection," *Insight*, vol. 44, no. 6, pp. 353-358, 2002.

[3] P. Cawley et al., "Practical long range guided wave testing: applications to pipes and rail," *Materials Evaluation*, vol. 61, 2003.

[4] S. Coccia et al., "Numerical and experimental study of guided waves for detection of defects in the rail head," *NDT & E International*, vol. 44, no. 1, pp. 93-100, 2011.

[5] P. Rizzo et al., "Ultrasonic guided waves-based monitoring of rail head: laboratory and field tests," *Advances in Civil Engineering*, 2010, doi:10.1155/2010/291293.

[6] Ministry of Railways, Government of India, Lok Sabha Unstarred Question No. 2 (rail fractures, 2008-09), answered 19 November 2009.

[7] Official documentation: scikit-learn and Espressif ESP32 / ESP-NOW.